24 **C**

E and B_1 form a velocity selector: $B_1Qv = QE \Rightarrow v = \dfrac{E}{B_1}$

So, ions P and Q have the same speed upon entering uniform magnetic field B_2

Inside B_2, both ions travel in circular paths. Magnetic force provides the centripetal force:

$$B_2 Q v = \frac{mv^2}{r}$$

$$\Rightarrow r = \frac{mv}{QB_2} = \frac{mE}{QB_1 B_2}$$

Since E, B_1, B_2 are constants, we have

$$r_Q = r_P \left(\frac{m_Q}{m_P} \right) \left(\frac{Q_P}{Q_Q} \right) = 3.7 \left(\frac{1}{1.5} \right) \left(\frac{2}{1} \right) = 4.9 \text{ cm}$$

25 **A**